

Arnav Singh

☎ +91 8825111505 | ✉ singharnav7763809@gmail.com

🌐 LinkedIn | 🐙 GitHub | 🐦 Twitter

EDUCATION

Pilani

Birla Institute of Technology and Science (BITS), Pilani
Concurrent specialization in Computer Science

Bachelor of Science (B.Sc.) in Computer Science

2025 – 2028

TECHNICAL SKILLS

- **Languages:** Java, Python, JavaScript, TypeScript, SQL
- **Backend & Systems:** Node.js, REST APIs, Microservices, WebSockets
- **AI / ML:** Machine Learning, Generative AI, LLM Integration, RAG Pipelines
- **Frontend:** React.js, Tailwind CSS, State Management
- **Cloud & DevOps:** AWS, Google Cloud Platform (GCP)

PROJECTS

Zenith-Collab (Real-Time Distributed System) 📄

React.js, Node.js, WebSockets

- Designed a real-time collaborative editor reducing multi-user editing conflicts using Operational Transformation.
- Engineered distributed synchronization logic ensuring consistency across concurrent editing sessions.
- Reduced real-time synchronization latency by **40%** using optimized WebSocket pipelines.

Profit Pilot (AWS Agentic AI Automation) 📄

Python, AWS, AI Agents

- Built an AWS-based autonomous agent system automating Tier-1 IT monitoring workflows.
- Designed agents parsing financial and operational data streams with **95% accuracy**.
- Reduced manual monitoring workload through automated alert-driven workflow execution.

TruthLens (Misinformation Detection Pipeline) 📄

Python, Deep Learning, RAG

- Built an LLM + RAG misinformation detection pipeline enabling credibility verification against trusted sources.
- Designed reliability scoring workflows by cross-referencing claims with verified knowledge repositories.
- Architected a modular backend supporting concurrent verification requests.

MediScan (OCR + Healthcare Data Extraction) 📄

Computer Vision, OCR, NLP

- Built an OCR + NLP pipeline converting handwritten prescriptions into structured medical JSON.
- Improved healthcare record digitization speed by **3x**, reducing manual entry errors.

PRODUCT IMPACT

- Reduced collaboration latency by **40%** in multi-user editing environments.
- Automated Tier-1 IT monitoring workflows using agent-triggered execution pipelines.
- Improved healthcare record digitization efficiency by **3x** using OCR + NLP extraction.
- Designed misinformation verification workflows improving reliability scoring using RAG.

ACHIEVEMENTS & HACKATHONS

Gen AI Exchange Hackathon (Google Cloud)

2025

- Built a GenAI legal-document simplification system using Vertex AI.

SuperHack 2025 (AWS & SuperOps)

2025

- Built an agentic AI system for IT management automation using AWS infrastructure.

OpenAI Academy x NxtWave Buildathon

Regional 2025

- Developed an AI-driven application using OpenAI models during a 2-day sprint.

Hack4Health (IIIT Delhi)

Finalist — 2025

- Rapidly prototyped a healthcare MVP under tight hackathon timelines.

POSITIONS OF RESPONSIBILITY

Developer Relations (DevRel) — Devnovate

- Supported developer onboarding and ecosystem engagement through technical sessions.
- Assisted workshops promoting adoption of developer tools in student communities.

Organizer — Hack Club Game Jam Bengaluru

- Organized a city-level Game Jam enabling student builders to prototype interactive projects.

Core Member — Entrepreneurship Cell

- Organized startup workshops and speaker sessions with industry mentors.